

Jenil Bhavinbhai Patel

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PROFESSIONAL SUMMARY

Software Developer with 3+ years of experience building performance-critical desktop, industrial and embedded software in C/C++, Qt 5/6 and QML. At Digital Shark Technology, develop an enterprise Video Management System (VMS), a SCADA/HMI platform with 15+ industrial protocol drivers, and a biometric iris-scanner HMI. Strong in multithreading, real-time telemetry, gRPC/Protobuf integration, hardware SDK integration, and CMake/CI-CD packaging. Background in automotive and embedded protocols (CAN, UART, SPI, I2C) on ARM microcontrollers.

TECHNICAL SKILLS

Languages: C++, C, Embedded C, QML, JavaScript, Python, Java, CAPL Scripting

Frameworks & Libraries: Qt 5.15/6 (QML, Qt Quick, Widgets, Multimedia, SerialPort, Qt gRPC/Protobuf, Qt SQL), OpenCV, FFmpeg, GStreamer, Android SDK

Protocols & Communication: gRPC, Protobuf, RTSP, ONVIF, Modbus (TCP/RTU), OPC UA, MQTT, CAN, UART, SPI, I2C

Databases & Security: SQLite, SQLCipher (encrypted storage), PostgreSQL, JWT authentication

Build, DevOps & Testing: CMake, Azure DevOps CI/CD, Docker, Git, CTest, windeployqt, Inno Setup, ApplImage, ARM cross-compilation, Keil uVision5, Proteus 8

Concepts: Multithreading, Real-Time Systems, SCADA/HMI, Video Streaming, Object-Oriented Programming, Data Structures, Hardware-Software Integration, Embedded Linux

Platforms & Tools: Linux (Ubuntu), Windows, Qt Creator, Visual Studio, VS Code, Android Studio, LPC2129 (ARM7)

WORK EXPERIENCE

Software Developer (Qt/QML | C++), Digital Shark Technology Pvt. Ltd.

Aug 2025 - Present

Bengaluru, Karnataka

Hi-Focus VMS Platform | Qt 6, QML, C++, gRPC, Protobuf, FFmpeg, GStreamer, RTSP, ONVIF, SQLCipher

- Developed features for a cross-platform enterprise **Video Management System** desktop client in **Qt 6/QML and C++**: multi-camera live view, recording schedules, timeline playback, PTZ control, ONVIF device integration, two-way audio talkback, alarm management and telemetry.
- Integrated the client with backend services using **Qt gRPC/Protobuf** and **MediaMTX RTSP** streams for device management, role-based user management, alarm records and AI alert search.
- Optimized multi-channel video decoding and rendering with **FFmpeg/GStreamer** worker threads, TurboJPEG decoding and GPU management to keep feeds smooth on constrained hardware.
- Implemented **SQLCipher**-encrypted storage for credentials and metadata, plus client-side security controls such as IP filtering and screen protection.

SharkView Platform (SCADA/HMI) | Qt 5.15/6, C++, QML, Modbus, PLC Tags, CMake, Azure DevOps

- Contributed to an enterprise **SCADA/HMI platform** (Studio IDE and Runtime) supporting 15+ industrial protocols, ISA-18.2 alarm management, SQLite historian, redundancy/failover and embedded JavaScript scripting.
- Implemented fieldbus and substation protocol drivers (**Modbus, PLC tags**) as Qt plugins for real-time telemetry acquisition into the runtime tag engine.
- Engineered reusable **QML** industrial visual components (tanks, pumps, valves, trends, 3D digital-twin widgets) bound to live tag data.
- Streamlined **CMake** build, ARM Linux cross-compilation toolchains and automated packaging/deployment across **Azure DevOps** CI/CD pipelines (Windows installer, Linux ApplImage, CTest suites).

Visiontek-HMI, Iris Scanner Platform | Qt 6, QML, C++, OpenCV, IDDK SDK, CMake, Inno Setup

- Engineered a high-speed HMI in **Qt 6/QML and C++** with **OpenCV**, interfaced with biometric iris-scanner hardware through the **IDDK SDK** over USB/UART for real-time video capture and image export (K1/K3/K7, JPEG2000).
- Implemented frame-capture loops, dynamic iris quality scoring and usable-area evaluation, streaming live frames to QML through a custom **QQuickImageProvider**.
- Built device lifecycle management: connection monitoring, capture retry and error recovery, and session timeout/warning handling.
- Designed a centralized **.env-driven multi-theme branding engine** propagated across C++, QML, Windows resources, CMake and installer scripts; automated one-step build, windeployqt staging, Inno Setup installer and SHA-256/MD5 checksums.

Embedded Systems Trainee, Vector India Private Limited

Nov 2024 - Aug 2025

Bengaluru, Karnataka

- Completed advanced training in embedded systems: microcontroller programming, RTOS concepts and automotive protocol integration.
- Built hands-on projects on the **LPC2129 (ARM7)** with hardware interfacing over **UART, I2C, SPI and CAN**.
- Applied C, C++, Embedded C and **CAPL scripting** for automotive ECU testing and network simulation; used Linux for cross-compilation and debugging.

Android Application Developer, Invica Infotech

Sep 2023 - Aug 2024

Surat, Gujarat

- Built a cross-platform Poster Maker mobile app in **React Native** with customizable templates, drag-and-drop tools and layer management.
- Delivered responsive UI with reusable components; enabled high-resolution export and social sharing.

Android Developer Intern, Prime Apps Infotech

Dec 2022 - May 2023

Ahmedabad, Gujarat

- Developed a real-time chat application in **Java** with text messaging, photo sharing, message reactions, biometric login and group chats.

EDUCATION

B.Tech in Computer Engineering, Dharmsinh Desai University, Nadiad, Gujarat

2019 - 2023

CGPA: 8.07 / 10

ADDITIONAL INFORMATION

Languages: English (Professional), Hindi, Gujarati